Technical Report
Rion Imaging

Team: 31 | **Date:** February 2026

1 Executive Summary

Rion Imaging proposes a 24-satellite CubeSat constellation—20 imaging satellites at 250 km and 4 infrastructure relays at 100 km—to achieve daily global coverage of the lunar surface. The architecture leverages projected advances in nanocomposite radiation shielding and solid-state batteries to overcome the extreme conditions of the lunar environment. Deployment via SpaceX Starship rideshare and controlled end-of-life disposal establishes a sustainable, commercially viable lunar observation capability.

2 Constellation

Our constellation architecture employs a layered CubeSat system designed to achieve daily global lunar surface coverage while maintaining robustness to spacecraft failure and a credible end-of-life (EOL) strategy. The constellation comprises 24 satellites, divided into a primary imaging layer and a secondary infrastructure layer.

The primary imaging layer consists of 20 identical CubeSats in near-polar (90°) circular low lunar orbits at an altitude of 250 km. Near-polar inclination ensures access to all lunar latitudes, while the satellites are distributed across 20 distinct orbital planes (18° RAAN spacing) to provide longitudinal diversity and reduce time to global coverage. With a fixed 3.667° aperture, the imaging payload achieves a swath width of approximately 16 km at this altitude. First-pass calculations indicate that around ≈ 220 strips are required for full surface coverage, while the constellation generates ≈ 218 strips per day, supporting daily global coverage with operational margin.

A secondary infrastructure layer of 4 CubeSats operates at ≈ 100 km altitude in selected highly stable inclination candidates ($\approx 27°$, 50°, 76°, 86°), providing store-and-forward communications between satellites as well as any client ground stations, operational resilience, and support for controlled EOL disposal. The constellation is deliberately sized with redundancy such that the loss of individual spacecraft does not prevent global coverage, and no satellite is intended to remain indefinitely in lunar orbit.[1]

3 Radiation

In lunar orbit, annual radiation levels from Galactic Cosmic Rays (GCR) and Solar Particle Events (SPEs) exceed 100 kRad, making shielding with an areal density of around 8–11 g/cm^2 necessary to limit them to an effective 15 kRad and maintain acceptable component tolerance. Due to the U2 CubeSats' size, only 8–10% of their total mass is available for radiation protection. Conventional aluminium walls (0.55–0.69 g/cm^2) are inadequate and generate harmful secondary radiation. [2]

With continuous innovation it is likely that by 2035, we will be using Hydrogen-Rich Nanocomposites made of MDPE (Medium-density polyethylene) with graphene oxide as reinforcement, maintaining high hydrogen shielding efficiency while providing mechanical reinforcement and thermal conductivity for lunar extremes.

Table 1: Reference Performance Metrics: Current vs. Future

Performance Parameter	Current Baseline	2035 Projection	2045 Estimate
Areal Density (g/cm^2)	0.55–0.69	6–8	7.53
Thickness (mm)	0.5–1.0	5–7	8
GCR Attenuation	13–15%	85–92%	92–95%
SPE Elimination	$<$5%	95–97%	98–99%
Mission Dose Tolerance	30–40 kRad	150–200 kRad	$>$300 kRad
Mass Budget (g)	—	180–220	210–230

By 2045, multiple technologies will combine to form fully graded, multifunctional hulls with structural panels that serve as both load-bearing and radiation-shielding components. A combination of MDPE–graphene nanocomposites with embedded boron-rich phases delivers an effective areal density of 8–10 g/cm^2 within the same 200–240 g mass budget.[3], [4], [5]

4 Battery

With the currently available technologies, using a U2 CubeSat in a lunar orbit with 45–60-minute eclipse times is infeasible due to temperatures cycling from –150°C to +150°C (a 300°C swing per 2-hour orbit). Assuming an average power draw of 35–50 W, it requires 150–200 Wh capacity in <300 g mass—demanding >500 Wh/kg specific energy. As such, advances in battery technology are necessary.[6], [7]

Table 2: Battery Technology Comparison

Chemistry	Wh/kg	Cycles	Rad (kRad)	Temp (°C)	2045 Status
Li-ion	120–150	3.5K–5K	20–30	–20 to +60	Obsolete
Li-ion HV	100–140	2K–3K	100–150	–40 to +85	Limited
Li-S	350–450	1.5K–2.5K	50–75	–30 to +80	Transition
SSB Li-metal	400–650	3.5K–25K	150–300+	–100 to +150	Deployable

SSB specs: 400–450 Wh/kg now; 550–650 Wh/kg by 2045. Cycles: 3.5K–5K electrochemical; 15K–25K thermal by 2045.

Using solid-state batteries with lithium-metal anodes and solid electrolytes should achieve the required performance numbers, as shown in 2. Solid electrolytes eliminate liquid-phase ionisation degradation, achieving 10–15× the radiation tolerance of Li-S and 150–300× that of Li-ion. Operation spans –100°C to +150°C (interface stability via ceramic coatings).[8], [9], [10], [11], [12], [13], [14]

5 Launch

Constellation deployment will be carried by ride-share on SpaceX Starship for easy deployment and incorporating its already planned integration with Lunar Gateway. Assuming a satellite mass of 2-3 kg per unit, a batch of 10 satellites will be deployed per launch. Additionally, the back up option as accounted in our costs estimates is a set of independent Electron launches to cislunar space.

6 End of Life

Although lifetime modelling has not been performed at this stage due to its inherent complexity in lunar orbit, a reliable estimate of the operational lifetime of both constellation layers will be produced prior to launch using high-fidelity orbital propagation. The secondary infrastructure layer is placed in maintenance-efficient long-lived “frozen” orbits and is therefore expected to exhibit a longer natural lifetime and lower station-keeping demand than the primary near-polar imaging layer.

The imaging layer is treated as a bounded-lifetime operational layer, intentionally designed to be retired earlier than the infrastructure layer. Prior to disposal, imaging spacecraft are transitioned into end-of-mission modes to maximise remaining value, including communications relay, calibration, and science demonstrations. Spacecraft are then passivated and removed from lunar orbit via controlled impact into non-sensitive corridors, avoiding polar volatile regions, heritage sites, and protected scientific areas. This approach mitigates the accumulation of long-lived debris in lunar orbit.

The infrastructure satellites are longer-lived, easier to track, and therefore suitable candidates for a selective sustainability demonstration using an experimental manoeuvring vehicle. These spacecraft are designed with modular interfaces to enable partial in-orbit component reuse where feasible (e.g. batteries, reaction wheels, communications hardware, and data storage). Following selective recovery, remaining spacecraft structures are disposed of via controlled impact into the same non-sensitive corridors. The recovered spacecraft would be brought back to gateway for flight back down to earth for data gathering and reuse.

7 Thermal Management

For satellites in lunar polar orbit, in an environment devoid of an atmosphere and where convective heat transfer is negligible, thermal control is largely governed by radiative and conductive mechanisms. Considering the volume and power constraints of a 2U CubeSat platform, passive radiative heat transfer is the primary means of thermal regulation.

In addition to the external surface coatings on the CubeSat’s aluminium walls, multi-layer insulation is applied to non-radiating surfaces to minimise heat loss during eclipse phases. Internally, thermal straps and heat-spreading structures can balance temperature gradients between subsystems. Active thermal control is limited to survival heaters for sensitive payload and avionics, with their power demands needing to be taken into account with the power budget. [15]

8 Hardware

Each 2U CubeSat (of approximately 2-3 kg mass) will comprise a high-resolution visible-light imaging system for lunar surface mapping and infrastructure monitoring, coupled with an infrared imaging system to survey shadowed regions and identify thermally anomalous features of interest such as polar cold traps. [16]

Attitude control of the CubeSat is achieved by a three-axis stabilised ADCS that incorporates reaction wheels, sun sensors, star trackers, and inertial measurement units. This would ensure the required pointing accuracy to minimise motion blur during imaging operations.

Communications and data handling are managed via the onboard computer and a high-gain antenna, with the Lunar Gateway serving as a secure relay node in order to reduce Earth-facing link budget requirements (as well as supporting secure data transmission for commercial and governmental clients).

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